ParaThink: Shared Context, Parallel Thinking and Expression

Context-mediated coordination / Independent turn policy / One playback owner

Incoming user audio

Nemotron Streaming ASR
Online transcript + prefix version

Transcript updates

Currently Available Dialogue Context + Runtime State
Transcript + generated history / Layer 2 CoT / XTurnix decisions
Backchannel timing suggestions / Layer 1 output / Listen–Speak state
Interruption events / generated, submitted, canceled and played records

Read / Actions

XTurnix Turn Policy
Listen: keep / start
Speak: keep / stop
All turn decisions

Event-driven turn decisions
On ASR updates / future CoT writes

Read / Timing

Layer 0 Backchannel Timing
System backchannel
Timing prediction

System backchannel

Read / Output

Layer 1 Expression
Generate spoken text
Request thinking
Resume with new input

Speech text

Request

Read / CoT

Layer 2 Async Thinking
Publish CoT only
No turn-action advice
Update one active task

CoT shared with XTurnix + Layer 1
Layer 1 reads at its next input boundary

Text FIFO
Submitted speech text / system backchannel, in order
Chunks carry output_id + chunk_id; accept active output only

Consume in order

start / stop control
Bypass Text FIFO
No Layer 1 wait

Output Runtime: Streaming TTS + Player
Single arbitration point / single owner of audible output
Priority stop: halt audio + cancel pending output
Reject late text / audio carrying an invalid output_id

Audio heard by the user: never revoked

Actual Listen / Speak state + interruption events

On a valid stop: execution in the same output runtime

1. Close old output_id
Block stale submissions

2. Halt + cancel
Cancel queued / buffered output
Discard late text / audio

3. Stop acknowledged
Publish Listen + interruption
Keep generated history

4. Wait for valid start
Fresh output_id + new text
Old queue stays canceled

ASR updates trigger XTurnix; future Layer 2 CoT writes also trigger a decision, never playback directly.

CoT-conditioned XTurnix requires training / Layer 2 CoT is internal, never playable text